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BUSINESS INTELLIGENCE

INDUSTRY TALK REPORT

Title of Talk:

Bridging Data & Decisions: The Role of Data Governance and Data Management in Modern Business Intelligence

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1.0 Introduction

On 20 May 2026, the Faculty of Computing Universiti Teknologi Malaysia hosted an insightful industry talk titled "Bridging Data & Decisions: The Role of Data Governance and Data Management in Modern Business Intelligence". The session was delivered by Mr. Faiz Fablillah, a highly experienced Senior Principal of Enterprise Data & AI Governance at PIDM and a certified professional with DAMA International. The two-hour talk discussed how modern organizations can move beyond simply handling data to making trusted data-driven decisions through proper data governance and management. In today's business environment, companies rely heavily on Business Intelligence (BI) and Analytics, technical knowledge alone is insufficient without strong governance and strategic data management.

Mr Faiz also highlighted the gap between textbook learning and the complex reality of real-world data ecosystems. He explained how organizations often face integration issues, missing documentation, and inconsistent data definitions before accurate analytics can be produced. This discussion was highly relevant to our Business Intelligence course as it emphasized that analytics tools can only perform well with high-quality and properly governed data.

2.0 Key Takeaways

As organizations grow, they can no longer rely on general software developers for complex data workflows. This is where the analytics engineer comes in that working right at the intersection of raw data infrastructure and business logic. Their primary responsibilities include building pipelines with organizing modern storage systems like data lakes for raw data, lake houses that combining storage and computing power, and data mesh setups for decentralized department ownership. Moreover, they integrate data by structuring steady data flows using ETL/ELT pipelines for live updates, and real-time streaming. Furthermore, analytics engineers need to treat data as a product by turning random operational data into a structured asset with clear owners, proper documentation, and strict quality guarantees.

In the real world, companies are dealing with hundreds of different source systems, broken data feeds, and conflicting definitions between departments. Without proper analytics and management, companies fall into a state called "Beautiful Ignorance." Analytics bridges this gap by turning that messy, unverified raw data into a single source of truth. It resolves the common corporate nightmare where the sales team and the finance team show up to a board meeting with completely different revenue figures. By implementing strict data quality checks, mapping out data lineage, and ensuring proper cataloging, analytics ensures that when a business leader looks at a KPI on a dashboard, they can fully trust it to make decisions.

Managing data manually is structurally impossible at enterprise scale, where environments regularly exceed ten thousand of data sources and tables. Modern data strategy relies on a combination of advanced governance tools and formalized institutional practices. Governance is standardized around the 11-pillar DAMA-DMBOK framework, which mandates the orchestration of Master Data Management (MDM) to establish a unified "Golden Record" of truth, alongside robust data quality frameworks to monitor data accuracy, completeness, and consistency across all analytical layers. To achieve true data observability, organizations leverage comprehensive metadata management paired with end-to-end data lineage tracing to visually map out data transformation lifecycles from origin to reporting. Security measures such as Role-Based Access Control (RBAC) and explicit data sensitivity classifications further protect these institutional assets. Furthermore, organizations are rapidly adopting agentic automation tools utilizing machine learning for automated profiling, anomaly detection, automated cataloging, and automated Personally Identifiable Information (PII) detection to ensure dynamic compliance with regulations. These automated systems operate under a "Human-in-the-Loop" paradigm to ensure that while machine intelligence executes high-volume operations, human oversight remains strictly accountable for policy design and risk judgment.

To show these concepts in action, there's three practical case studies shared. First, a bank faced issue in conflicting customer data across its apps, ATMs, and CRM systems which led to serious errors like sending loan offers to deceased individuals. By adopting MDM, they removed duplicates and created a single source of truth. This fixed their reporting errors and reduced regulatory risk by 40%. Similarly, a telecom provider overcame a trust crisis where employees rejected over 200 dashboards by utilizing data catalogs and lineage tracing to audit pipelines. This enables them to triple the employee adoption and reduce the system to 50 validated dashboards. A Healthcare AI project mentioned where researchers needed quick access to sensitive patient files but manual PDPA compliance checks created delay and errors. They implemented an automated governance system using AI to instantly detect and mask private personal details while maintaining an unchangeable audit trail. This allowed them to securely share data much faster with zero data breaches.

3.0 Reflection and Critical Insights

One of the most valuable insights gained from this talk is that data governance is equally important as data engineering in modern Business Intelligence (BI). In academic settings, emphasis is often placed on building efficient and structured data pipelines. However, the speaker, Mr. Faiz, highlighted the gap between the “textbook pipeline” and the “reality pipeline.” In real-world scenarios, organizations frequently deal with messy data, missing documentation, and unclear data ownership.

These challenges significantly affect business outcomes. Poor data quality can result in inaccurate dashboards, ultimately leading to incorrect decisions. This highlights that technical expertise alone is insufficient. Effective communication and collaboration are also essential, as alignment across departments is required to resolve inconsistencies and establish common data definitions.

The talk also demonstrated strong connections to concepts learned in the course, particularly metadata and master data management (MDM). While metadata is defined as “data about data”, its practical importance is reflected through the use of a data catalog, which functions as a centralized inventory of data assets. With a well-implemented data catalog, users can easily discover, understand, and utilize data, enabling both technical and non-technical stakeholders to make informed decisions. In addition, the explanation of MDM through the “Golden Record” concept provided a clear real-world application. Master data, such as customer information, must be standardized across departments. Without MDM, conflicting figures may arise, leading to confusion and inefficiency. By establishing a single, trusted version of the data, consistency can be achieved and decision-making can be improved.

From a broader perspective, the importance of analytics engineering in the current business ecosystem is evident. Many organizations still face challenges with underutilized data, and analytics engineers play a crucial role in transforming raw data into reliable insights. The transition from manual processes to automated and AI-driven approaches further enhances efficiency. For example, AI tools can detect data quality issues early, preventing inaccurate information from reaching dashboards and decision-makers.

Several questions are also raised for further exploration. For instance, how can strong data governance be implemented without reducing organizational flexibility? What are the most effective strategies for resolving conflicting data definitions across departments? Additionally, to what extent can AI support data governance while still requiring human oversight?

Overall, this session broadens the understanding that successful BI implementation requires not only technical skills, but also strong governance, communication, and critical thinking.

4.0 Conclusion

In short, this industry talk provided our group with an eye-opening reality check on the actual progress of business intelligence. After hearing Mr. Faiz break down the complexities of company's data spaces made us realize that success in this field involves much more than just writing flawless code, cleaning data and designing beautiful dashboards. The talk clearly emphasized that without robust data governance and reliable management frameworks, a company's analytical tools lose their worth entirely. It taught us that data must be carefully guarded, catalogued and protected as a valuable company’s product before it can ever be converted into a reliable guide for business decisions.

On an academic level and for our future career path, this talk has shifted our perspective significantly. As students pursuing our Data Engineering major, it is incredibly easy to get trapped in the mindset that building efficient pipelines is the final goal. This talk showed us that true expertise requires bridging that technical data engineering foundation with strategic, critical thinking for data governance too. Moving forward into our final semesters in campus and going for internships in two more months, the feelings become much more motivated to look beyond textbook scenarios. We intend to focus not just on data pipeline, storage and integration but also on mastering the governance practices, documentation and automated tools that turn real-world, messy data into trustworthy business insights.